

Emre CANCIOGLU

OPERATIONAL TECHNOLOGIES SUPERVISOR · INDUSTRIAL DIGITALIZATION & AI

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8+ Years Experience | 3 Facilities · 2-Person Team | 57% Cost Reduction | 3 Publications + 6 Awards

Summary

Results-driven Operational Technologies Supervisor with over 8 years of experience leading digital transformation and applied-AI initiatives on the industrial production floor. Currently overseeing OT digitalization and AI strategy across **3 manufacturing facilities**, managing a 2-person engineering team and coordinating across **10+ cross-functional departments**. Delivers measurable impact through **predictive maintenance**, **computer-vision-based safety systems**, **GenAI-powered automation**, and **energy & sustainability analytics**, consistently translating shop-floor data into operational efficiency, cost savings, and safer working conditions. Deep hands-on expertise with **PLC, SCADA**, and industrial protocols (**OPC UA, MQTT, Profinet**), balanced with strong software/cloud and leadership capability (**Node.js, React, Azure/AWS, Kubernetes**). Recognized with multiple 1st-place finishes and an Innovation Special Award across İnci Holding's group companies.

Skills

OT Systems	PLC (Siemens, Allen-Bradley, Mitsubishi) SCADA DCS HMI MES OPC UA MQTT Modbus (TCP/RTU) Profinet EtherNet/IP CC-Link Kepware Unified Namespace Digital Twin ISA-95 NAT VLAN Segmentation Industrial Firewalls IEC 62443
AI & Data	Predictive Maintenance Computer Vision Machine Learning Deep Learning GenAI/LLM Automation (RAG, n8n, Copilot, GPT, Claude) MLOps Edge Computing ETL/Stream Processing SPC Grafana Power BI Data Analytics
Leadership	Team Leadership & Mentoring Cross-functional Collaboration Project & Budget Management Bias for Action
DevOps	Azure AWS Docker Kubernetes Rancher Terraform Jenkins GitHub Actions NGINX MS IIS PM2 Git
Development	Node.js Express.js Django FastAPI REST API Kafka React Redux JavaScript TypeScript Python MS SQL MongoDB PostgreSQL DynamoDB Elasticsearch
Languages	Turkish (Native) English (Professional)

Work Experiences

İNCİ GS YUASA/İNCİ AKÜ

OPERATIONAL TECHNOLOGIES SUPERVISOR — 3 PRODUCTION FACILITIES, TÜRKİYE

Jan 2026 - Present

- Leading OT digitalization and applied-AI strategy across **3 production facilities**; managing a 2-person engineering team and coordinating across **10+** cross-functional departments. Delivered **15+** process-improvement projects since taking on the role.
- Scaled the OT data integration backbone to **300+** managed data channels and database connections (up from ~200) across **PLC, SCADA, MES**, and historian systems — integrating **36** new production lines/machines in 2026 alone (furnaces, mills, assembly, filling, and welding stations) and building a self-service **OPC UA**-based alerting platform that lets any user configure real-time custom alarms across all connected machines.
- Grew the applied-AI portfolio to **4** active predictive-maintenance/quality models and **25** AI-driven automation projects, including RAG-based knowledge systems and GenAI/multi-agent workflows (**n8n**, Microsoft Copilot, GPT, Claude) that cut day-long reporting cycles down to seconds.
- Deployed an AI-based human and forklift detection safety system, real-time air-quality/particulate monitoring in high-exposure areas (contributing to measurable reductions in workers' blood lead levels), and real-time anomaly detection on production/utility systems that helped prevent a potential fire incident — part of an expanding compliance program (tracked audit actions grown from 3 to 7 annually) sustaining **zero workplace accidents for 4 consecutive years**.
- Leading R&D and innovation visibility for the OT function — coordinating a **TÜBİTAK**-funded R&D project and preparing **6** internal/external innovation-competition submissions annually, contributing to multiple 1st-place and Innovation Special Award wins.

SENIOR OPERATIONAL TECHNOLOGIES (OT) ENGINEER — MANISA, TÜRKİYE

Dec 2022 - Dec 2025

- Accomplished **19%** increase in operational efficiency by reducing production cycle times, as measured by improved production throughput, through the implementation of a **Unified Namespace (UNS)** based on **ISA95** standards.
- Achieved **14%** improvement in data reliability by enhancing real-time data flow through the infrastructure optimization with **OPC UA, MQTT**, and **Kafka**, which also led to a 90% improvement in the response time to customer complaints.
- Reduced production monitoring delays by **32%**, as indicated by faster identification of production anomalies, by designing real-time dashboards in **Grafana** and **Node.js** for lead-acid battery manufacturing.
- Led cost-saving digital transformation initiatives, achieving a **57%** reduction in operational expenses, as reflected by decreased overhead costs, by modernizing **OT/ICS** systems and integrating automation technologies.

DVD VALVES — MANISA, TÜRKİYE

- Led a TEYDEB project (\$200k funding) managing embedded systems and **web service** integration for smart irrigation hydrants, using **Node.js** and **React**, resulting in **22%** improvement in reporting and accounting services efficiency.
- Developed control systems for irrigation, improving efficiency by **18%** using **DCS**, as demonstrated by optimized system response times.
- Led design and development of a control system for hydraulic butterfly valves in large-scale international water projects (**Italy, Dubai, Colombia**), reducing project costs by **30%** and saving **\$80k**.

SOFTWARE DEVELOPMENT ENGINEER

Jul 2019 - Dec 2021

- Integrated **IIoT** systems with the **Focas** library, improving CNC monitoring accuracy by **48%**, as evidenced by enhanced real-time visibility.
- Optimized **OEE** analysis tools, increasing efficiency by **12%**, by analyzing signals from **CNC, PLC, and web services**, leading to minimized machine idle times.

PART-TIME SOFTWARE DEVELOPMENT ENGINEER

Jul 2018 - Jul 2019

- Achieved **3x faster production cycle** by automating the welding process with **PLC** and **HMI**, as realized by shortened cycle time.

DIMES

İzmir, Türkiye

PART-TIME PROJECT ENGINEER

Jul 2017 - Jun 2018

- Enhanced efficiency by **4%** through real-time monitoring with **PLC** and **IIoT**, as validated by the reduction in maintenance needs.

Projects

Selected signature initiatives among numerous projects delivered across roles.

Energy Monitoring, Forecasting & Anomaly Detection Platform

Manisa, Türkiye

PROJECT LEADER

2025 – Present

- Retrofitted **200+** electricity, water, and natural-gas meters to remote-reading, built the underlying database, and launched an energy dashboard visualizing historical machine-level consumption, per-unit-product energy use, and carbon footprint — integrated with **SAP** and preparing the facility for **ISO 50001** certification.
- Built a compressed-air leak anomaly-detection algorithm fusing compressor pressure data, real-time **MES** production signals, and **SAP** machine-level production-confirmation data, enabling early leak intervention and reducing compressor energy consumption.
- Added ML-based energy forecasting driven by planned production volumes (**XGBoost**, LightGBM, GBM regression, R^2 up to **0.944**), co-authored as peer-reviewed research at the 13th International European Conference on Interdisciplinary Scientific Research (Tirana, Albania, May 2026).

Maestro

Manisa, Türkiye

PROJECT LEADER

Dec 2023 – Present

- Built a **Unified Namespace (UNS)** unifying **MES, SAP, PLC, IIoT**, and third-party data across the plant, leading a **20-person** team; stood up new OT network infrastructure (NAT, switches, firewalls) across **3 facilities** with an **RBAC/LDAP**-secured architecture, enhancing traceability by **40%**.
- Delivered end-to-end visibility via **Power BI** and custom dashboards showing real-time machine data, per-unit test results, and real-time **SPC**-based anomaly alerts — powering the digital twin and Digital Product Passport; accelerated reporting by **60%** using **Kafka, Node.js**, and **React**, saving **\$156K** annually.
- Integrated two **YOLO**-based computer-vision safety models: **AI-ODS**, which auto-stops robots when a person enters their work envelope, and **AI-FDS**, which issues hologram-style warnings to prevent forklift-pedestrian and forklift-forklift collisions at blind spots — contributing to zero workplace accidents.
- Added predictive maintenance using vibration and temperature sensors, with **SAP**-linked equipment lifecycle tracking.

Smart Irrigation Hydrant System (TÜBİTAK TEYDEB 1501)

Manisa, Türkiye

PROJECT LEADER

Jan 2022 – Nov 2022

- As project lead overseeing the **~\$3.5M TÜBİTAK TEYDEB 1501** budget, led a **9+**-engineer team building the embedded software and desk-top/mobile applications for the pre-charge and ultrasonic metering modules of this smart irrigation hydrant system, cutting unit cost by **~68%** versus the outsourced solution and bringing after-sales technical support fully in-house.

Honors & Awards

Cevdet İnci Incentive Awards are held across İnci Holding's group companies (5 sister companies, 9+ facilities); İnci GS Yuasa Stars is company-specific.

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|------|---|-------------------|
| 2025 | Finalist , Sustainability Business Awards (Technology & AI Category) | İstanbul, Türkiye |
| 2025 | 1st Place , Cevdet İnci Incentive Awards | İzmir, Türkiye |
| 2025 | 3rd Place , İnci GS Yuasa Stars | Manisa, Türkiye |
| 2024 | 1st Place , Cevdet İnci Incentive Awards | İzmir, Türkiye |
| 2024 | Innovation Special Award , Cevdet İnci Incentive Awards | İzmir, Türkiye |
| 2024 | 1st Place , İnci GS Yuasa Stars | Manisa, Türkiye |
| 2023 | 1st Place , Cevdet İnci Incentive Awards | İzmir, Türkiye |

Presentations & Publications

13th International European Conference on Interdisciplinary Scientific Research

Tirana, Albania

WRITER | RESEARCHER

May 2026

- Developed ML regression models (XGBoost, LightGBM, GBM) forecasting electricity and compressed air consumption on battery production lines, achieving R^2 up to 0.944 for short-term forecasts.

R&D & Innovation 2024

Manisa, Türkiye

WRITER | RESEARCHER

Dec 2024

- Data Integration for Industry 4.0 and IIoT: Unified Namespace-Based Digital Transformation with MQTT, OPC UA, and Node.js.

Human-Computer Interaction Optimization and Robotic Applications

Türkiye

WRITER | RESEARCHER

Jun 2021

- Fault Detection and Diagnosis in Process Control Systems using Machine Learning (ensemble learning) Methods from Poincaré Plot Measurements, improving classification accuracy to 89.5% on the Tennessee Eastman Process.

International Medical Device Conference

Antalya, Türkiye

WRITER | RESEARCHER

Sep 2020

- Long-Short Term Memory (LSTM)**-Based Heart Sounds Analysis and Classification: built an LSTM model classifying phonocardiogram recordings into Normal, Murmur, Extrasystole, and Artifact categories, achieving 79.0% accuracy (Journal of Intelligent Systems and Applications, Vol. 3(1), 2020).

Certifications

41 NORTH BUSINESS SCHOOL

"GELECEĞİ SAHİPLENİYORUZ" BLACKBOX BUSINESS CHALLENGE™

Jun 2026

(Credential ID: INC_2026_0406_4118)

"GELECEĞİ SAHİPLENİYORUZ" CİDDİ OYUNLAR™ İLE GELECEĞİ SAHİPLENEN İNOVASYON PROJELERİ

Jul 2026

(Credential ID: INC_2026_2606_4106)

KALMER KALİTE YÖNETİM MERKEZİ

"ISO 9001:2015 & IATF 16949:2016" TRAINING

Feb 2023

(Document No: 39530)

N-SİSTEM

ISO 9001:2015, ISO 14001:2015, ISO 45001:2018

Nov 2021

(Credential ID: 2021/3994)

MANİSA YUNUSEMRE HALK EĞİTİM MERKEZİ

HYDRAULICS & PNEUMATICS DEVELOPMENT AND ADAPTATION TRAINING

Feb 2020

(64 hours | Document No: 75476720200065196147)

DATABASE MANAGEMENT – SQL DEVELOPMENT AND ADAPTATION TRAINING

Dec 2019

(56 hours | Document No: 75476720190063875731)

İZMİR HALK EĞİTİM MERKEZİ

COMPUTER-AIDED DESIGN (AUTOCAD)

Dec 2017

(160 hours, İzmir Büyükşehir Belediyesi | Document No: 16420920180053334601)

PROGRAMMABLE LOGIC CONTROL (PLC)

Dec 2017

(296 hours, İzmir Büyükşehir Belediyesi | Document No: 16420920170052946539)

Educations

İzmir Katip Çelebi University

İzmir, Türkiye

MSC. IN ELECTRICAL & ELECTRONICS ENGINEERING

Sep 2019 - Jun 2022

GPA: 3.64/4.00 (100% English-Medium Program).

Coursework: Statistical Process Control | Data Acquisition and Control | Applied Machine Learning | Artificial Neural Networks.

Thesis: Fault Detection and Diagnosis of the Tennessee Eastman Process Using Statistical Analysis and Machine Learning Methods from Poincaré Plot Measurements.

İzmir Katip Çelebi University

İzmir, Türkiye

BSC. IN ELECTRICAL & ELECTRONICS ENGINEERING

Sep 2014 - Jul 2019

GPA: 3.14/4.00 (100% English-Medium Program).

Coursework: Process Control and Instrumentation | Optimization in Engineering | Industrial Automation | Signals and Systems.

Thesis: Design of a Cylindrical Welding and Polishing Machine with PLC-Based HMI Control.